

The catch-basin classifier, explained honestly.

What we train, what's in the training set, why we train it at all, and how we keep the accuracy number defensible. This is the answer to "isn't Claude doing the work?"

1 · The gap we fill

Cyvl grades pavement, signs, and markings — but **every physical asset has condition: null** (verified across all asset types). All 381 Somerville catch basins have a location and a street-level photo, and no condition score. The NVIDIA track asks teams to *train a model on Cyvl data*; we train the model that fills exactly this gap: catch-basin condition (clear / clogged / damaged) from the photo.

2 · What is actually in the training set

Element	What it is
Input (X)	The catch-basin photos — pixels, or in the fast path each photo's CLIP/DINOv2 feature vector
Target (y)	The label Claude Vision assigned to that photo: clear / clogged / damaged / uncertain (uncertain dropped from training)
What the model learns	A function <code>image → label</code> — it adjusts its weights to reproduce Claude's labels from the image

At **inference** the model is told nothing — it looks at a photo and predicts the label itself. Claude is the **teacher** (labels once); the small model is the **student** (learns to imitate). This is knowledge distillation.

3 · Why train a model at all, if Claude already gives the label?

Fair question — and the one a judge will ask. Two honest answers:

- **Scale and cost.** A frontier-model API call per image is fine for 381 basins. It is not fine for hundreds of thousands of assets across a city, rescanned every year. The distilled student runs for cents per image, offline, indefinitely. *Claude labels once; the small model serves at city scale for free.*
- **The track requires a trained model on Cyvl data** — this satisfies it genuinely rather than bolting on something fake.

Stated plainly: at 381 basins the trained model is a **proof of the scaling pattern**, not something we couldn't do with Claude alone. We say that out loud — the value is that it generalises to 380,000 assets, and to every other Cyvl city.

4 · Keeping the accuracy number honest (the key move)

If we grade the student against Claude's labels, we only measure "how well it copies Claude" — and Claude's labels could themselves be wrong. So:

1. The team hand-verifies a held-out set of ~30–40 photos. **These human-checked labels are the ground truth.**
2. The student trains on Claude's labels, but is **scored against the human-verified set**, never against Claude.
3. We report that number, plus 1–2 real misclassifications, and we **abstain below a confidence threshold** (low-confidence renders "unverified" on the map — never a confident wrong colour).

Now "accuracy" means *how well the model matches humans* — a real, defensible metric, not a circular one.

5 · The pipeline, end to end

Step	Action
Gather	Pull all 381 catch basins via the Cyvl MCP; each record carries a CloudFront photo URL — download the crops (~5 min)
Auto-label	Claude Vision labels each photo (clear/clogged/damaged/uncertain), ~150 in minutes for a few dollars of the \$200 credit
Audit	Team verifies ~30–40 labels → this becomes the human ground-truth test set; record agreement %
Train	Fast path: CLIP/DINOv2 frozen features + logistic-regression head (scikit-learn, seconds, CPU). GPU path (the NVIDIA flex): fine-tune a small ViT with HuggingFace
Score + show	Classify all 381 → map layer; report accuracy vs the human set; abstain below confidence

Feeds the product, not bolted on: a clogged or damaged basin raises that street's dig-readiness score — the classifier is an input to the ranking, which is what "genuinely useful" means under the rubric.

Risk discipline: this is FIRST on the cut list. With ~120 training images, overfitting is real — the human-graded test set and the abstain threshold are what keep it credible. Default to the fast path; attempt the ViT fine-tune only if the ML owner is comfortable and time allows. A broken classifier scores worse than no classifier.

6 · The Q&A answer

"ISN'T CLAUDE DOING THE WORK?"

"Claude generates the training labels — it's the teacher. The model we train is a small student that reproduces those labels from the image, so it runs at city scale for cents instead of a frontier call per asset. We grade it against human-verified labels, not Claude's, so the number is honest. At 381 basins it proves the pattern; the point is it scales to 380,000 and to every Cyvl city."